PART III · THE GROUND UNDER THE CHORDS

Scales and keys

Notes travel in families. Measure the distance between two notes, build the scales that whole pieces are made from, then learn to read the key a piece lives in.

Intervals
the distance between notes

›

Scales
major and minor

›

Keys
signatures and the circle

IN THIS PART

Counting intervals 51
Major, minor and perfect 53
Major scales 55
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Keys and key signatures 59
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The circle of fifths 63
Tips: when a note sounds wrong 65

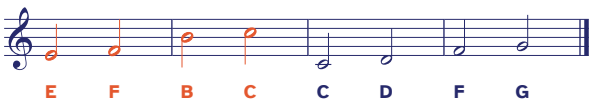
PART III · SCALES AND KEYS · 1

Counting intervals
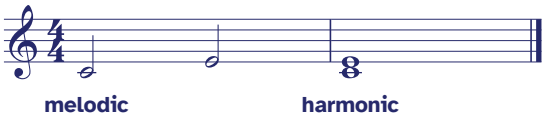
How far apart are two notes? That gap is an **interval**. The very next key is a **semitone**; two make a **tone**. To number it, count the letter names, both ends included.

THE SECRET

Count the letters, both ends included: C to E is a 3rd.

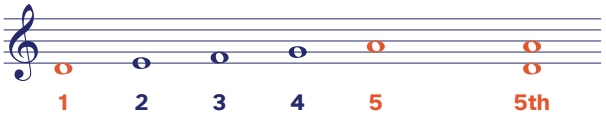


E–F and B–C are semitones; C–D and F–G are tones.



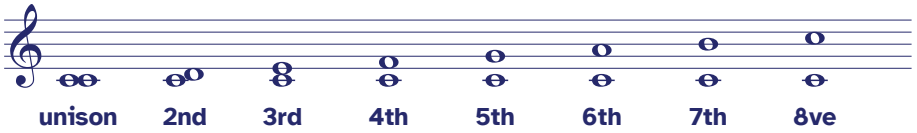
Melodic: one note after the other. Harmonic: together.

Counting the number



From **D** up to **A** (the red notes): D E F G A is five letters, so the interval is a 5th, whether the notes come one after the other or together.

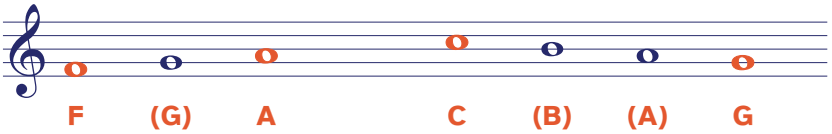
Interval numbers from C



Your hand is a ruler: thumb on C, little finger on G: five fingers on five white keys, a 5th.

EXAMPLE 1

How far apart are these notes?



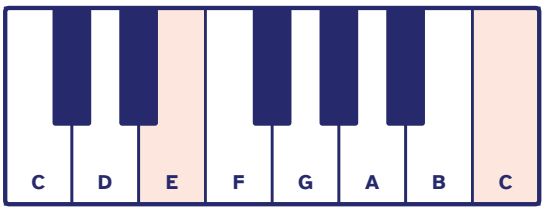
1. **Call the first note 1:** F is 1.
2. **Count every letter to the second note:** F G A is 3 letters: a 3rd.
3. **Going down works the same:** C B A G is 4 letters: a 4th.

F up to A is a **3rd**; C down to G is a **4th**. Sharps and flats never change the number.

PART III · SCALES AND KEYS · 1 · CONTINUED

EXAMPLE 2

Name an interval on the keyboard
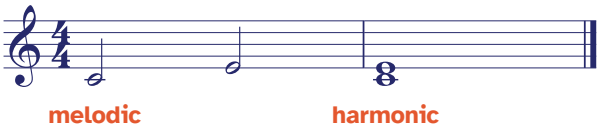


1. **Find both notes:** E and the C above it.
2. **Count the letters:** E F G A B C is six.
3. **Both ends count:** so the interval is a 6th.

E up to C is a **6th**.

EXAMPLE 3

Melodic or harmonic?
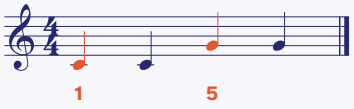


1. **The first bar:** C then E, one after the other.
2. **The second bar:** C and E together.
3. **Count both:** C D E, a 3rd each time.

The same 3rd: **melodic** when played one after the other, **harmonic** when played together.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



The leap from “twin-” to “kle” is C up to G: C D E F G, **five letters, a 5th**.

Try it. Sing the first two notes of any song, then count the letters between them.

THINK OF IT Monday to Wednesday is three days, not two: you count the day you start on. Intervals count the same way.

TRAP

- × Counting only the gaps, not the notes: always count both ends.
- × Letting a sharp or flat change the number: C to G♯ is still a 5th.

SEE ALSO

Major, minor and perfect p. 53 · Major scales p. 55

PRACTISE IT

Workbook Scales and keys 1 · Online: Library › Intervals

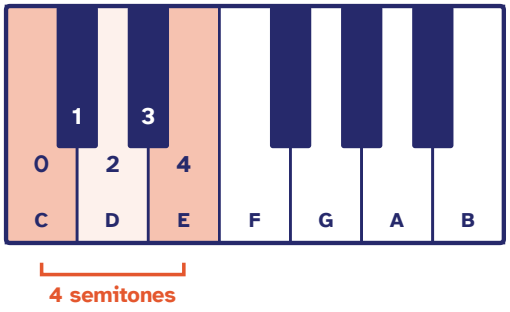
PART III · SCALES AND KEYS · 2

Major, minor and perfect

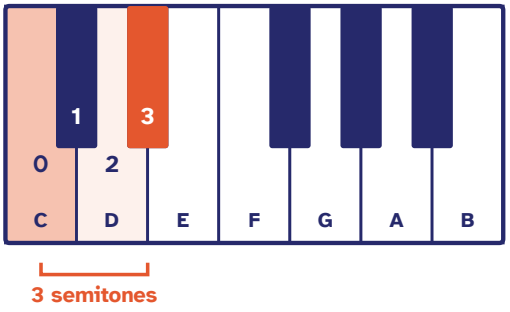
Two intervals with the same number can differ in size. C–E and C–E♭ are both 3rds, but one is a semitone wider. The **quality** (major, minor, perfect…) comes from counting **semitones**.

THE SECRET
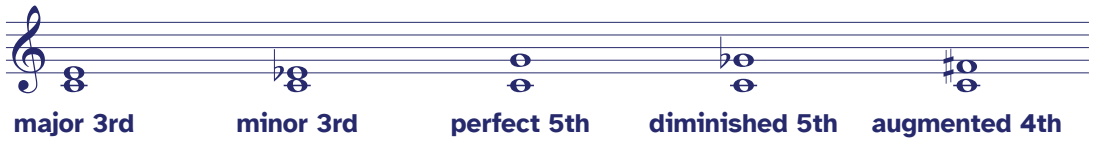
Count the semitones: 4 make a major 3rd, 3 a minor 3rd.



From **C** to **E**: count every key, black and white. 4 semitones: a major 3rd.



From **C** to **E♭**: 3 semitones, one fewer: a minor 3rd.



Size in semitones

NUMBER	SMALLER	MAIN	WIDER	FIRST MET
unison		0 (perfect)		Level 1
2nd	1 (minor)	2 (major)		Level 1
3rd	3 (minor)	4 (major)		Level ½
4th		5 (perfect)	6 (augmented)	Level 1
5th	6 (diminished)	7 (perfect)		Level 1
6th	8 (minor)	9 (major)		Level 2
7th	10 (minor)	11 (major)		Level 2
octave		12 (perfect)		Level 1

2nds, 3rds, 6ths, 7ths
diminished ‹ **minor** ‹ **major** ‹ augmented

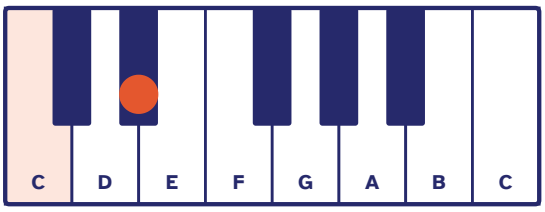
unisons, 4ths, 5ths, octaves
diminished ‹ **perfect** ‹ augmented

Each step on the ladder is one semitone. Six semitones (an augmented 4th or diminished 5th) is the tritone. Turned upside down, an interval's numbers add to 9 and major and minor swap: a major 3rd becomes a minor 6th.

PART III · SCALES AND KEYS · 2 · CONTINUED

EXAMPLE 1

Major or minor 3rd?

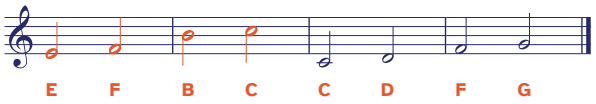


1. **Count the letters:** C D E is three, so it is a 3rd.
2. **Count the semitones:** C to C♯, D, E♭ is three.
3. **Three semitones** make a minor 3rd; four would be major.

C to E♭: a **minor 3rd**.

EXAMPLE 2

Tone or semitone?
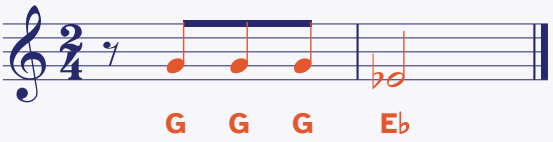


1. **E to F:** the very next key, no black key between: a semitone.
2. **C to D:** one key between (C♯): a tone.
3. **B to C** and **F to G**: check the same way.

E–F and B–C are semitones; C–D and F–G are tones.

YOU KNOW THIS

Beethoven, Symphony No. 5



Da-da-da-DUM: G down to **E♭** is four semitones, a **major 3rd**. Three semitones would be minor.

Try it. Count the keys from G down to E♭: F♯, F, E, E♭.

THINK OF IT A major and a minor 3rd are a smile and a frown: the same face, one small change at the corners.

TRAP

- × Counting white keys only: count every key, black and white.
- × Calling a 4th or 5th “major”: they are perfect.

SEE ALSO

Counting intervals p. 51 · Chords in a key p. 72

PRACTISE IT

Workbook Scales and keys 1 · Online: Library › Intervals
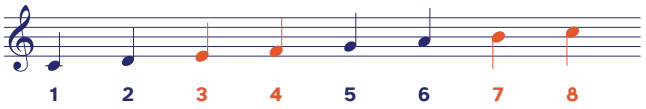
PART III · SCALES AND KEYS · 3

Major scales

Climb from a key note to the next one with the same name, following a fixed recipe of tones (T) and semitones (S): that is a **scale**. The recipe, not the starting note, makes it major or minor.

THE SECRET

Every major scale: tone, tone, semitone, tone, tone, tone, semitone.



C major: T T S T T T S. The semitones fall between 3–4 and 7–8.

SCALE	STEPS	FIRST MET
major	T T S T T T S	Level ½
natural minor	T S T T S T T	Level ½
harmonic minor	T S T T S T+S S	Level ½
melodic minor (up)	T S T T T T S; down as the natural minor	Level 2
chromatic	every semitone	Level 2
pentatonic (black keys)	T T T+S T T+S	Level 0

Every step has a name

1 tonic · **2** supertonic · **3** mediant · **4** subdominant · **5** dominant · **6** submediant · **7** leading note

Major or minor?

The same key note can start a major or a minor scale: C major and C minor. The pattern decides. Major sounds bright and settled; minor, darker.

HOW TO

Practise a scale

1. **Hands separately,** slowly, saying the finger numbers aloud.
2. **Fingering for C, G, D, A and E major** (right hand): 1 2 3, thumb under, 1 2 3 4 5.
3. **Listen for evenness:** every note the same length and loudness.
4. **Then hands together,** one octave, then two.

EXAMPLE 2

Build F major

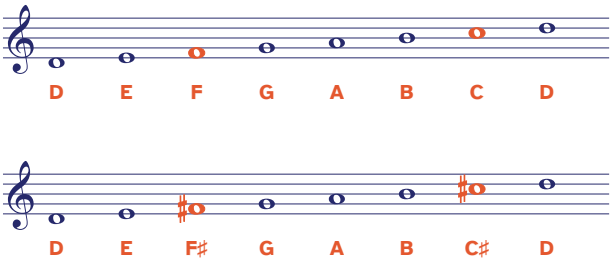
1. **Write the letters** from F to F: F G A B C D E F.
2. **Check the steps** against T T S T T T S: A to B is a tone, but it must be a semitone.
3. **Lower B** to B♭: now A to B♭ is a semitone and B♭ to C a tone.

F major: **F G A B♭ C D E F**, one flat.

PART III · SCALES AND KEYS · 3 · CONTINUED

EXAMPLE 1

Build D major
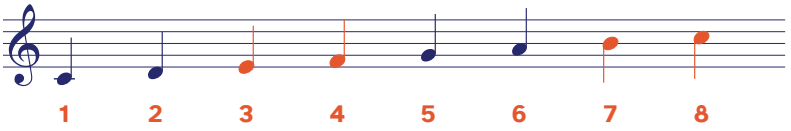


1. **Write the eight letters** from D to D, every letter once.
2. **Check the steps** against T T S T T T S: E–F is too small, B–C is too small.
3. **Raise F and C** a semitone to fix them.

D major: **D E F♯ G A B C♯ D**, two sharps.

EXAMPLE 3

Where are the semitones?
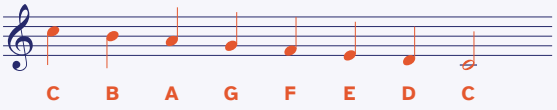


1. **Number the degrees** 1 to 8 from C.
2. **Look for no black key between:** E–F and B–C.
3. **Those are degrees 3–4 and 7–8.**

Every major scale has its semitones between **3–4** and **7–8**.

YOU KNOW THIS

Joy to the World



The first line is the **C major scale**, top to bottom: C B A G F E D C.

Try it. Play C major down from the top C and sing “Joy to the world” over it.

THINK OF IT A scale is a staircase with two short steps. Wherever you build it, the short steps sit in the same places: between 3 and 4, and between 7 and 8.

TRAP

- × Using the same letter twice: every letter appears once (F♯, not G♭, in G major).
- × Forgetting the semitone between 7 and 8: the leading note pulls up to the key note.

SEE ALSO
Minor scales p. 57 · Keys and key signatures p. 59

PRACTISE IT
Workbook Scales and keys 2 · Online: Library › Scales

PART III · SCALES AND KEYS · 4
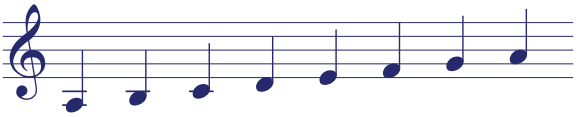
Minor scales

Every minor scale starts from the natural minor, the notes of its relative major from the 6th degree up. Two small changes give the harmonic and melodic forms, which sound stronger at the top.
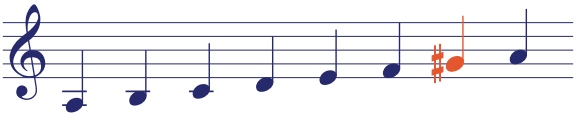
THE SECRET

Three minor scales, one home note: only the 6th and 7th change.

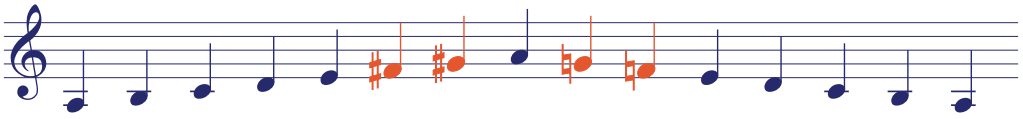
A minor in three forms



Natural: the notes of C major, from A to A. T S T T S T T.



Harmonic: the 7th raised a semitone (**G♯**). The gap F–G♯ is a tone and a half.



Melodic: the 6th and 7th raised going up (**F♯ G♯**), lowered coming down (**G♮ F♮**).

Relative minor

Same key signature, three semitones below the major: C major and A minor share every note.

Parallel minor

Same key note, different signature: C major (no sharps or flats) and C minor (three flats).

EXAMPLE 1

Build E harmonic minor

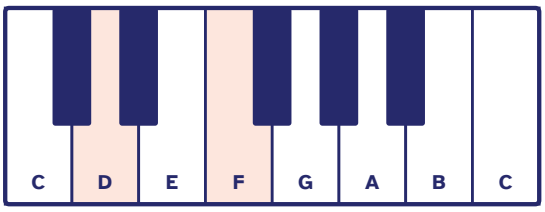
1. **Borrow G major's notes** starting on E: E F♯ G A B C D E.
2. **Raise the 7th:** D becomes D♯.
3. **Check the gap** C to D♯: a tone and a half.

E harmonic minor: **E F♯ G A B C D♯ E**.

PART III · SCALES AND KEYS · 4 · CONTINUED

EXAMPLE 2

Find the relative minor of F major
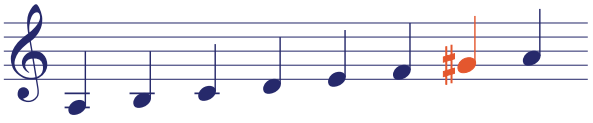


1. **Start on F,** the key note.
2. **Count three semitones down:** E, E♭, D.
3. **Check the signature:** D minor has one flat, B♭, like F major.

The relative minor of F major is **D minor**.

EXAMPLE 3

Natural or harmonic minor?
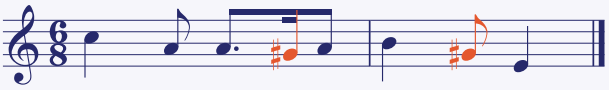


1. **Start on A** and read up to A.
2. **Look at the 7th:** G♯, raised.
3. **Nothing else changes** from the natural minor.

A raised 7th only: the **harmonic minor**.

YOU KNOW THIS

Greensleeves



A minor, with a G♯: the raised 7th of the harmonic minor, leaning hard up to A.

Try it. Play it once with G♯, once with G: hear the pull disappear.

THINK OF IT The relative minor is the same clothes worn by a different person: the same notes, a new key note. The parallel minor is the same person in different clothes: the same key note, some notes changed.

TRAP

- × Raising the 7th in the key signature: it is written as an accidental each time.
- × Playing the melodic minor's raised notes on the way down.

SEE ALSO
Major scales p. 55 · Keys and key signatures p. 59

PRACTISE IT
Workbook Scales and keys 2 · Online: Library › Scales

PART III · SCALES AND KEYS · 5

Keys and key signatures

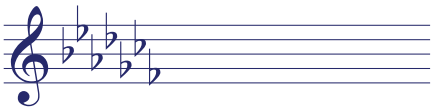
Why write F♯ fifty times? The **key signature** at the start of each line says it once, for the whole piece. Its sharps or flats always line up in the same order.

THE SECRET

The key signature says once, at the start, which notes are always sharp or flat.



Sharps: F C G D A E B. The key note is a semitone above the last sharp.

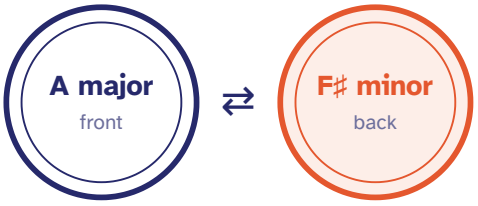


Flats: B E A D G C F. From two flats, the second-last flat names the key.

To remember the order: **F**ather **C**harles **G**oes **D**own **A**nd **E**nds **B**attle; backwards for the flats: **B**attle **E**nds **A**nd **D**own **G**oes **C**harles' **F**ather.

MAJOR	RELATIVE MINOR	SIGNATURE	FIRST MET
C major	A minor	—	Level ½
G major	E minor	1♯	Level ½
D major	B minor	2♯	Level 1
A major	F♯ minor	3♯	Level 3
E major	C♯ minor	4♯	Level 4
B major	G♯ minor	5♯	Level 4
F major	D minor	1♭	Level ½
B♭ major	G minor	2♭	Level 1
E♭ major	C minor	3♭	Level 3
A♭ major	F minor	4♭	Level 4
D♭ major	B♭ minor	5♭	Level 4

Six and seven sharps or flats (F♯, C♯, G♭, C♭ major and their minors) come at Levels 4–6.



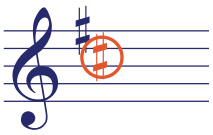
One signature, two keys

Every key signature belongs to **one major key and one minor key**, like the two sides of a coin: three sharps are A major or F♯ minor. The minor is the **relative minor**, three semitones below the major.

PART III · SCALES AND KEYS · 5 · CONTINUED

EXAMPLE 1

Name the keys
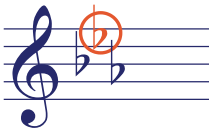


1. **Count:** two sharps, F♯ and C♯.
2. **The last sharp** is C♯; one semitone up is D.
3. **The relative minor,** three semitones below D, is B.

Two sharps: **D major** or **B minor**.

EXAMPLE 2

Read a flat key signature
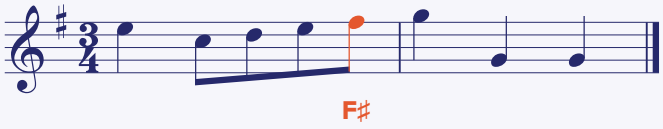


1. **Count the flats:** three, B♭ E♭ A♭.
2. **The second-last flat** is E♭: the key note.
3. **Its relative minor,** three semitones down, is C minor.

Three flats: **E♭ major** or **C minor**.

YOU KNOW THIS

Minuet in G



One sharp in the key signature: every F is **F♯**, like the one just before the top G. That sharp is why it is in G major.

Try it. Find every F in the piece and play it as F♯.

THINK OF IT A dress code on a party invitation: written once at the top, it applies to every guest all night. Only an accidental, a word to one guest, overrules it, and only until the bar line.

TRAP

- × Writing a key signature in the wrong order: sharps always F C G D A E B.
- × Forgetting that a signature names two keys: a major and its relative minor.
- × Reading it once and forgetting it: glance at the key signature at the start of every line.

SEE ALSO

Major scales p. 55 · Finding the key p. 61 · The circle of fifths p. 63

PRACTISE IT

Workbook Scales and keys 2 · Online: Library › Keys

PART III · SCALES AND KEYS · 6

Finding the key
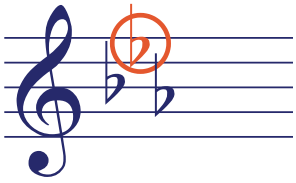
A key signature names two keys, a major and its relative minor. Two quick rules (and one to remember) find the major; one more finds the minor; then the music itself tells you which of the two it is.

THE SECRET

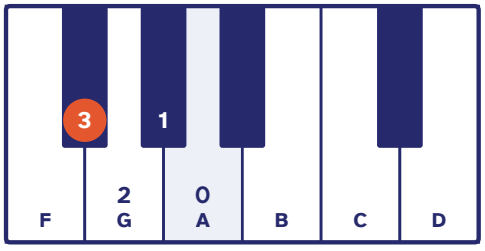
Last sharp, one step up: the major key. Last flat but one: the major key.



Sharps: the last sharp, then one semitone up.
Last sharp **G♯** → up a semitone → **A major**.



Flats: the second-last flat is the key.
B♭ **E♭** A♭ → **E♭ major**. One flat alone is F major.



Major to minor: three semitones down.
A, G♯, G, **F♯** → **F♯ minor**, the relative minor of A major (count every key, black and white).

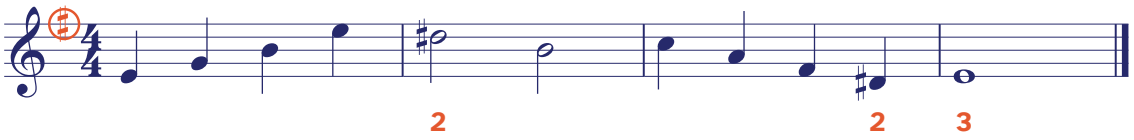
Minor keys and their clue notes

MINOR KEY	SIGNATURE	CLUE NOTE (RAISED 7TH)
A minor	none	**G♯**
E minor	1♯	**D♯**
D minor	1♭	**C♯**
B minor	2♯	**A♯**
G minor	2♭	**F♯**

PART III · SCALES AND KEYS · 6 · CONTINUED

EXAMPLE 1

What key is this tune in?

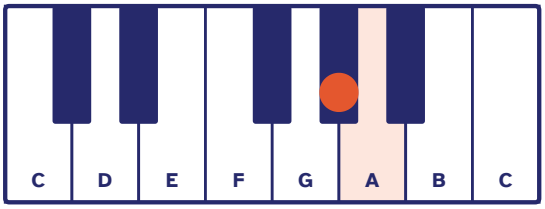


1. **The key signature:** one sharp, so G major or its relative, E minor.
2. **The clue notes:** D♯ keeps appearing. D♯ is the raised 7th of E minor; G major has no D♯.
3. **The last note:** E, the key note of E minor.

The key: **E minor**. Check it the same way with any piece you play.

EXAMPLE 2

Major or minor? No sharps or flats
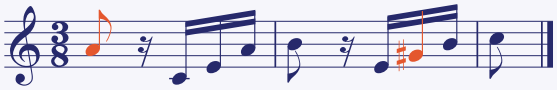


1. **The signature:** none, so C major or A minor.
2. **The clue notes:** G♯ appears. G♯ is the raised 7th of A minor.
3. **The last note:** A.

No sharps or flats, G♯ clues, ending on A: **A minor**.

YOU KNOW THIS

Für Elise (Beethoven)



No key signature, a **G♯** as the clue, and the music keeps coming home to **A**: **A minor**.

Try it. Play A, then G♯ A: hear the G♯ lean into home.

THINK OF IT Detective work: the key signature narrows it to two suspects; the clue note and the last note name the culprit.

SEE ALSO
Keys and key signatures p. 59 · The circle of fifths p. 63

PRACTISE IT
Workbook Scales and keys 2 · Online: Library › Keys

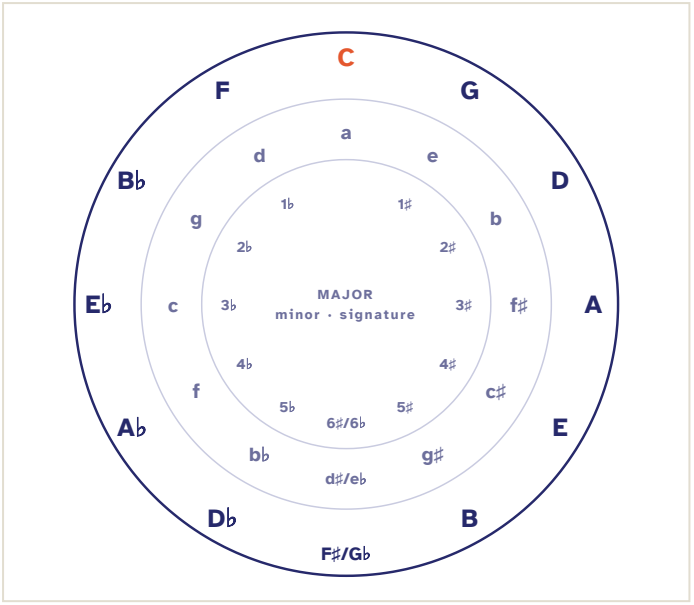
PART III · SCALES AND KEYS · 7

The circle of fifths

All the keys on one clock face. Going round clockwise, each key is a **5th higher** than the last and has **one more sharp** (or one fewer flat). Anticlockwise, each is a 4th higher, a 5th lower, with one more flat.

THE SECRET

Clockwise adds a sharp, anticlockwise adds a flat: every key on one clock.



How to read the clock

Outer ring: major keys. **Middle ring:** their relative minors (small letters). **Inner ring:** the key signature.

Neighbours are close

Keys next to each other share all but one note, so music moves between them easily: from C major to G major or F major, or to the relative minor. These are the **related keys**.

Six jobs for one circle

Key signatures
Count the steps from C: clockwise for sharps, anticlockwise for flats.

Relative minors
Look straight in from any major key.

The main chords
A key's two neighbours give its other main chords: in C, the chords of F and G.

Cadences
The strongest close, V to I, is one step anticlockwise: G to C.

Changing key
Pieces usually move one step round, or to the relative minor.

Scale practice
Go round the circle: one new sharp or flat each time.

The first steps round the circle

STEPS	CLOCKWISE (SHARPS)	ANTICLOCKWISE (FLATS)
1	G major: F♯	F major: B♭
2	D major: F♯ C♯	B♭ major: B♭ E♭
3	A major: F♯ C♯ G♯	E♭ major: B♭ E♭ A♭

PART III · SCALES AND KEYS · 7 · CONTINUED

C major's neighbours

KEY	WHERE ON THE CIRCLE	WHY IT MATTERS
G major	one step clockwise	the dominant: the strongest pull home
F major	one step anticlockwise	the subdominant: a gentle move away
A minor	straight in from C	the relative minor: the same notes
E minor, D minor	in from G and F	the relatives of the neighbours

EXAMPLE 1

Why do G major and D major sound so close?

1. **Find both:** D is one step clockwise from G.
2. **Compare the scales:** G major has C; D major has C♯.
3. **Count:** six notes the same, one different.

Neighbours share six of their seven notes, so music moves between them easily.

EXAMPLE 2

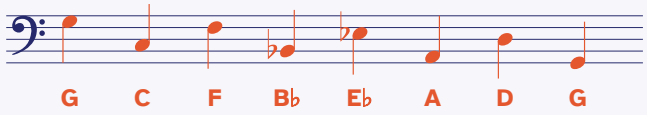
Find A major's key signature

1. **Start at C** at the top of the circle: no sharps.
2. **Go clockwise:** G, D, A is three steps.
3. **Three steps,** three sharps: F♯, C♯, G♯.

A major has **three sharps**.

YOU KNOW THIS

Passacaglia (Handel), the bass



Each bass note is a **5th below** the one before (or a 4th above): G C F B♭ E♭ A D G, one step round the circle each chord.

Try it. Play the bass line slowly and hear it pull towards the last G.

THINK OF IT Like a colour wheel: neighbours blend, opposites clash. C and G sit side by side; C and F♯ sit opposite.

TRAP

- × Going the wrong way round: clockwise adds sharps, anticlockwise adds flats.
- × Forgetting the relative minors on the middle ring.

SEE ALSO

Keys and key signatures p. 59 · Finding the key p. 61 · Chords in a key p. 72

PRACTISE IT

Workbook Scales and keys 2 · Online: Library › Related keys

PART III · SCALES AND KEYS · TIPS

When a note sounds wrong

You read the note right, yet it sounds sour. The slip is usually in the key, not the note. Name the slip first, then fix only that note.

TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
A note sounds sour, though you read it right	**the key signature**: is this note sharp or flat all the way through?	p. 59
Two notes sound too close or too far apart	**the interval**: count the letters, then the keys between	p. 51
A minor tune sounds wrong near the end	**the raised 7th**: harmonic and melodic minor sharpen it	p. 57
A scale sounds wrong in the middle	**the semitone steps**: 3–4 and 7–8 in every major scale	p. 55
A sharp is missing from the key signature	**the order of sharps**: F C G D A E B	p. 59
You are not sure which major key it is	**the last sharp**: the key is one note above it	p. 61
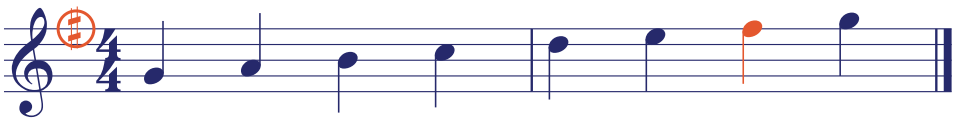
HOW TO

Three checks before you play

1. **The key signature:** say its sharps or flats aloud.
2. **The key:** name it from the last sharp or the last-but-one flat, then check the last note.
3. **The scale:** play it once, so your fingers know which black keys belong.

EXAMPLE

Find the slip



1. **Read the key signature:** one sharp, on the top line, F (ringed).
2. **The red note sits on that line**, so the sharp belongs to it too.
3. **No natural cancels it**, so this F is F♯, here and all through the piece.

Play **F♯**, the black key. Plain F is the most common slip in G major.

THINK OF IT Like proofreading: a spelling mistake jumps out once you know which words you usually get wrong. Keep a short list of your own slips, and check those first.